Chopper Interlock Module

Test Plan Document DRAFT

Implementation in LabVIEW FPGA

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SCOPE

This document defines the Test Plan for the Chopper Interlock Module. This module contains the FPGA configuration VI and its Project containing an EPICS IOC server and client. The audience of this document are the test engineers who will execute the Test Plan and produce the Test Report.

The test plan covers the following aspects:

- Test procedures for ensuring all requirements are verified.
- Performance tests.
- The tests are classified for the execution of the SAT and FAT.

TEST PROCEDURES

[CIM-GNRL-0000] LabVIEW version control

CIM-GNRL-0000	**LabVIEW version control**
Requirement(s)	
Description	
Pre-requisite	
Procedure	
1. Run NI MAX application. 2. [Checkpoint] Check NI MAX contains 3. LabVIEW 2014 SP1 f10 a. LabVIEW-RT 2014 b. LabVIEW DSC 2014 c. LabVIEW FPGA 2014 4. LabVIEW Xilinx vivado tools 13 5. NI-RIO 14.0.1	
Pass Criteria	
1. Software defined in Procedure steps contains the expected versions.	

[CIM-GNRL-0001] EPICS CA PV access

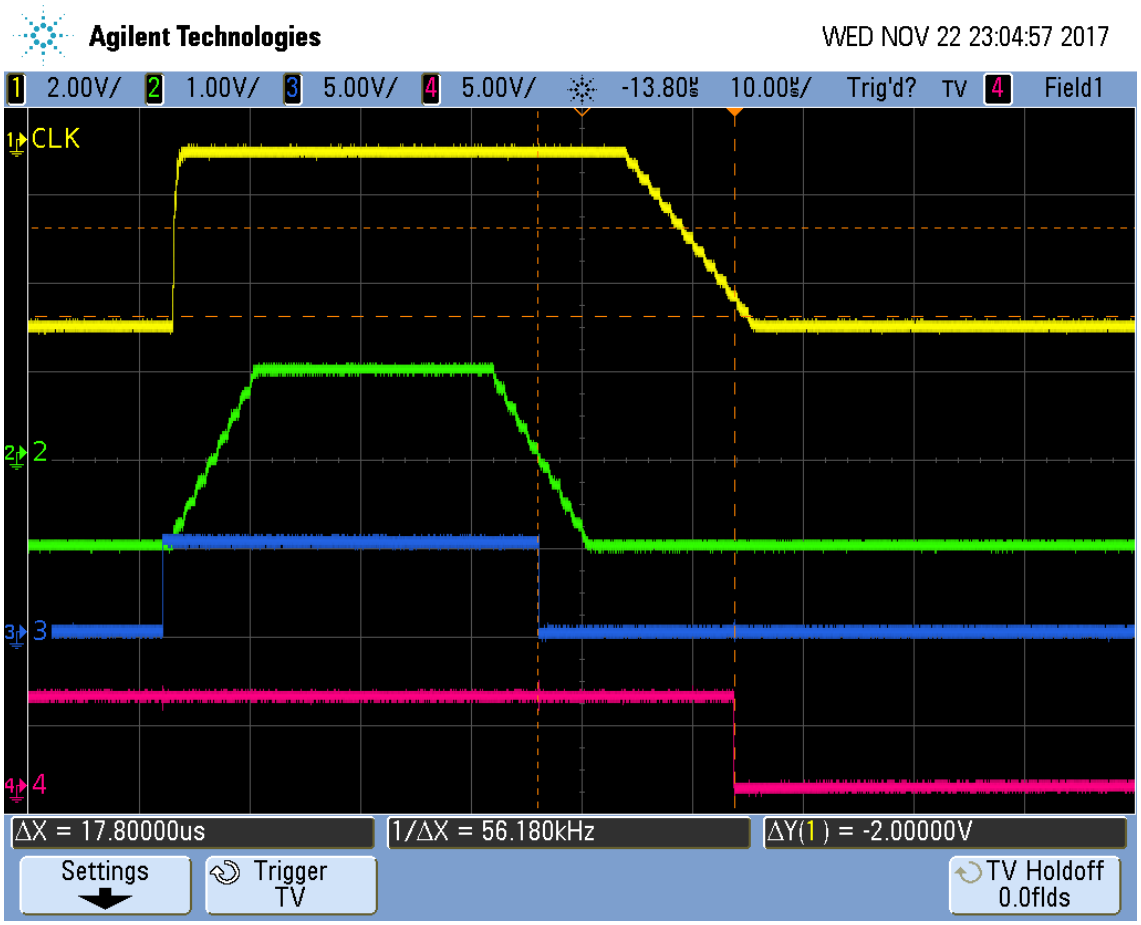
CIM-GNRL-0001	**EPICS CA PV access**
Requirement(s)	
Description	
Pre-requisite	
Procedure	
Pass Criteria	

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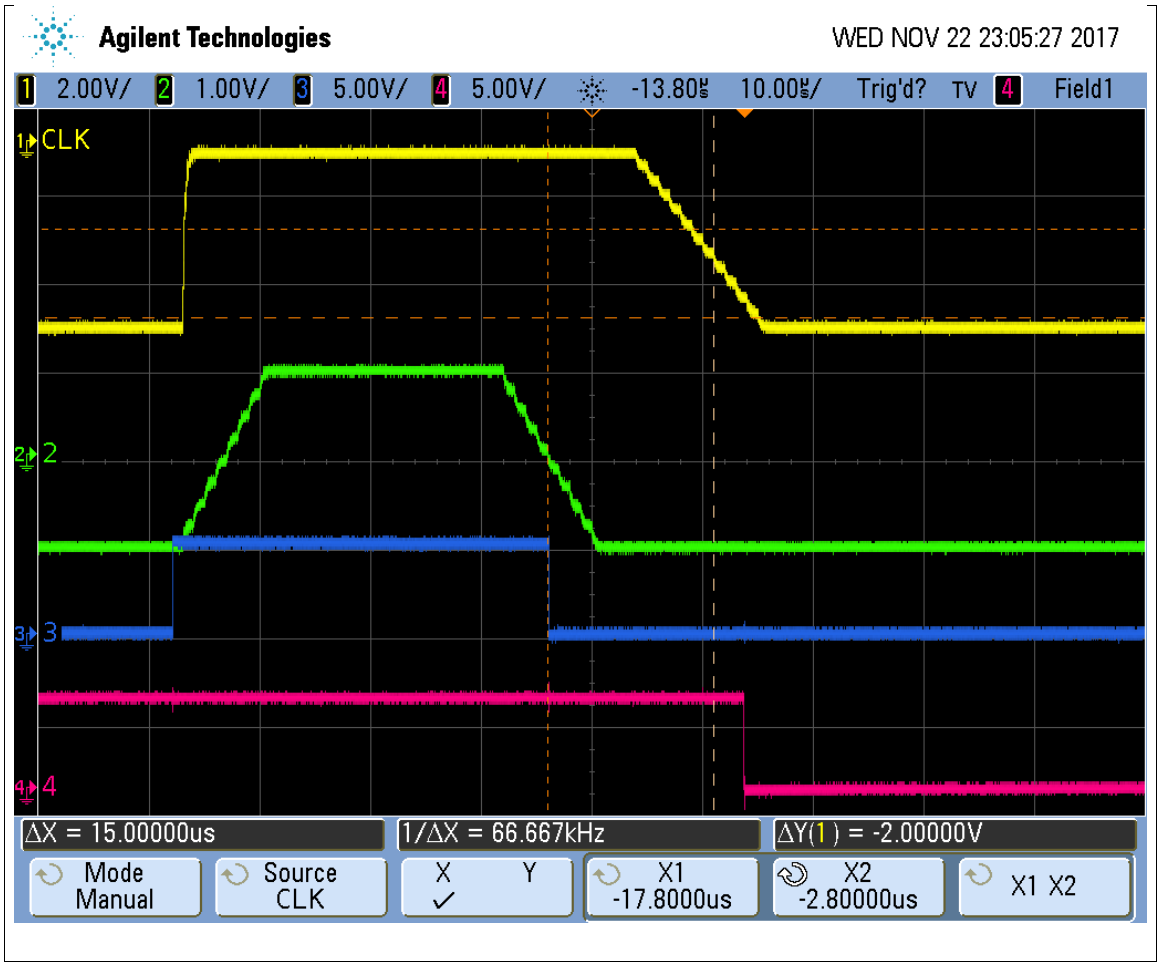
[CIM-FUNC-0000] CIM enable state test

CIM-FUNC-0000	CIM enable state test
	PASSED

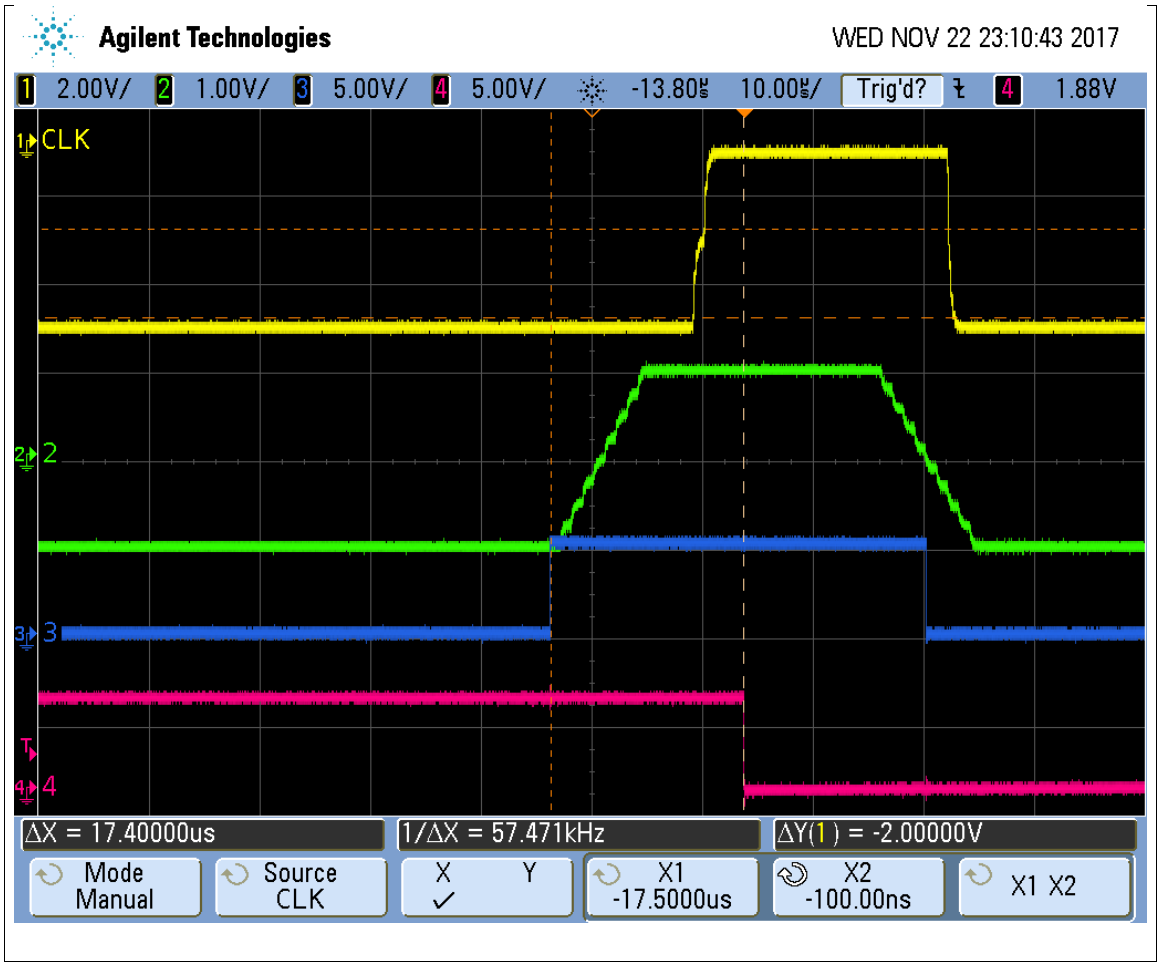
All combinations are not shown. A slight delay of 2us was detected on the VMON signal detection.



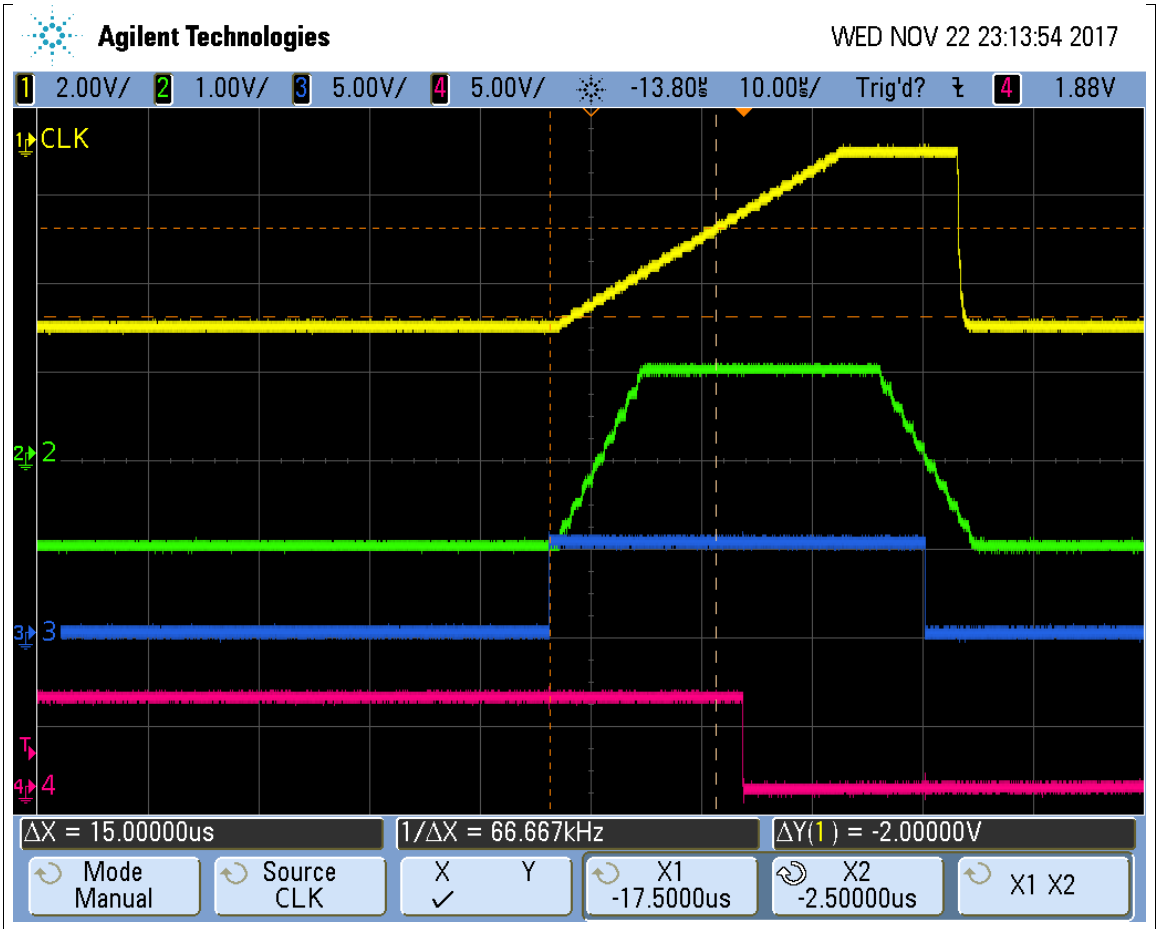
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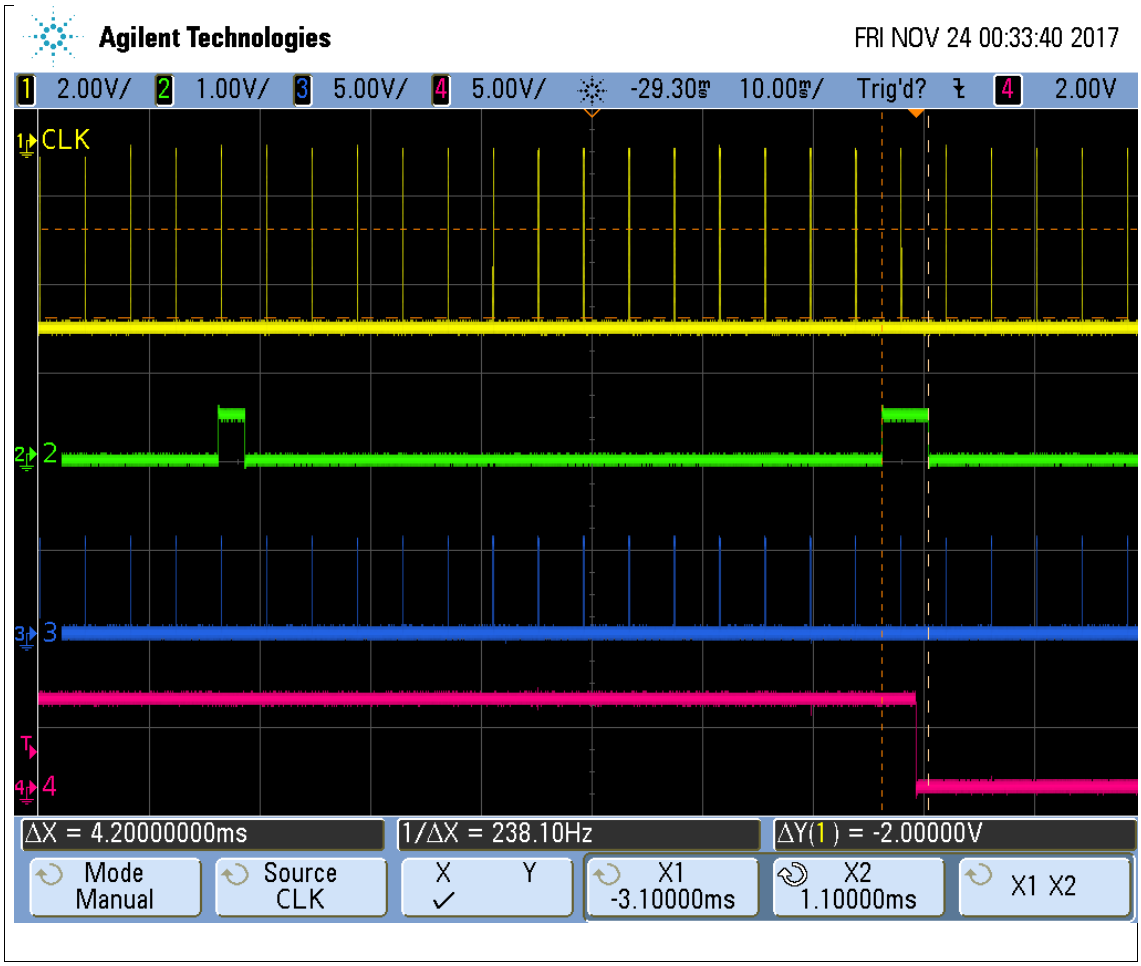


The delay is consistent with FPGA loop cycle and module read times. Can be adjusted through optimization.

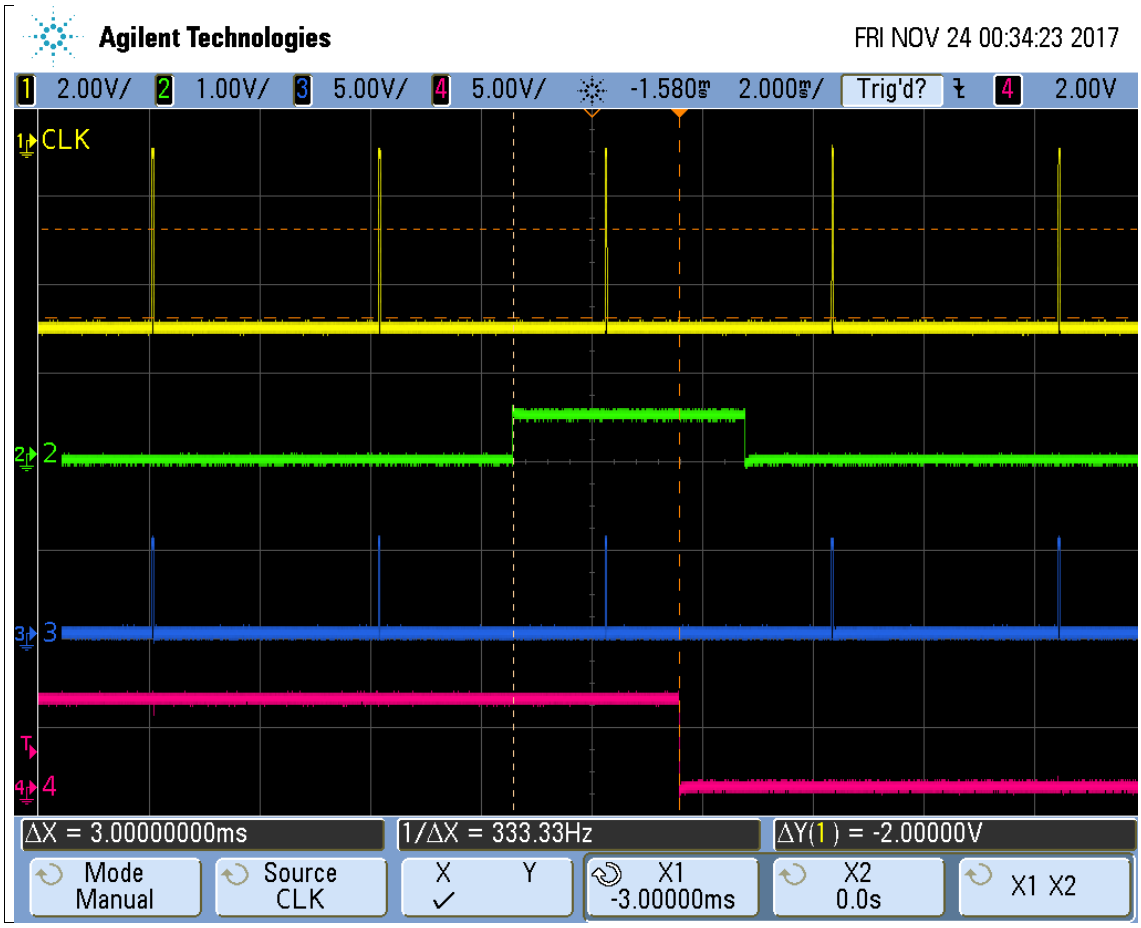
[CIM-FUNC-0001] CIM Duty Cycle monitoring

CIM-FUNC-0001	CIM Duty Cycle monitoring
	PASSED

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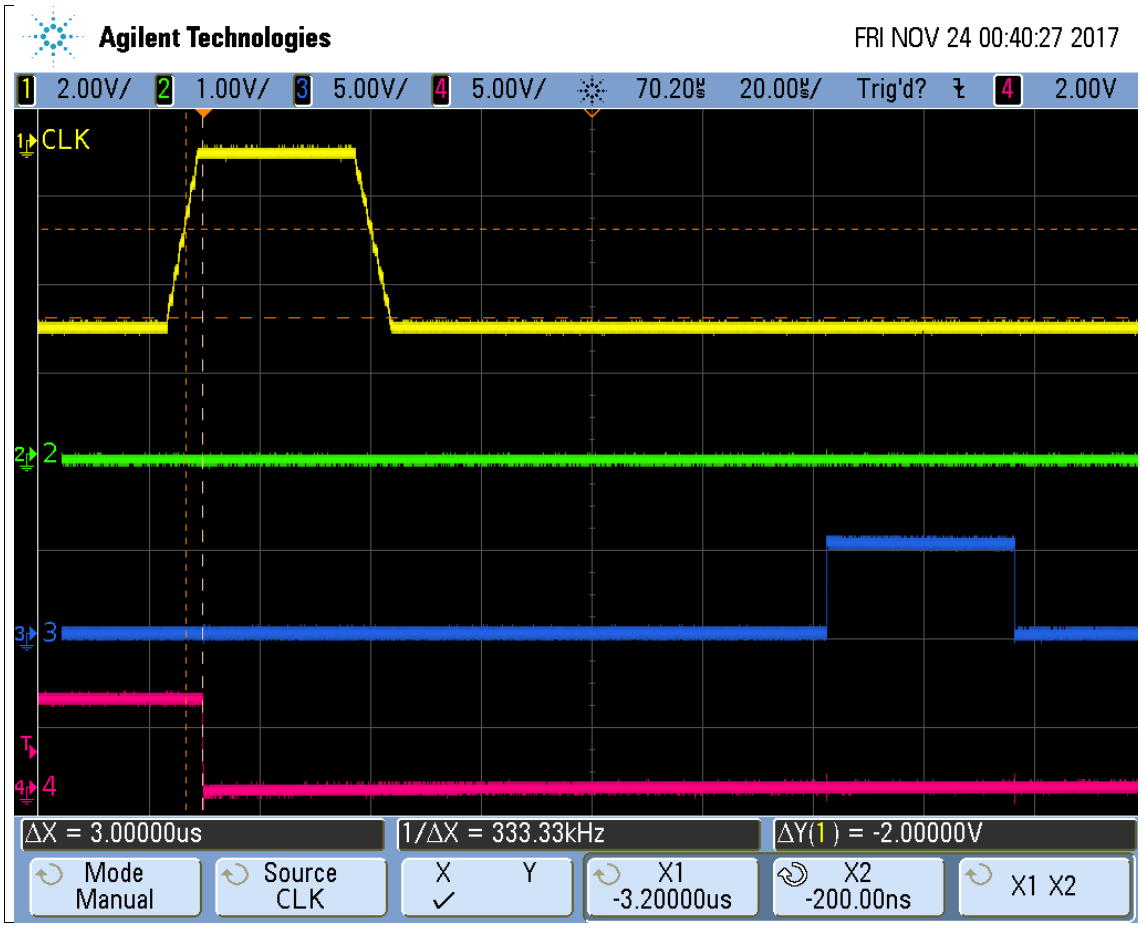
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[CIM-FUNC-0002] CIM GATE OFF state check

CIM-FUNC-0002	CIM GATE OFF state check
	PASSED

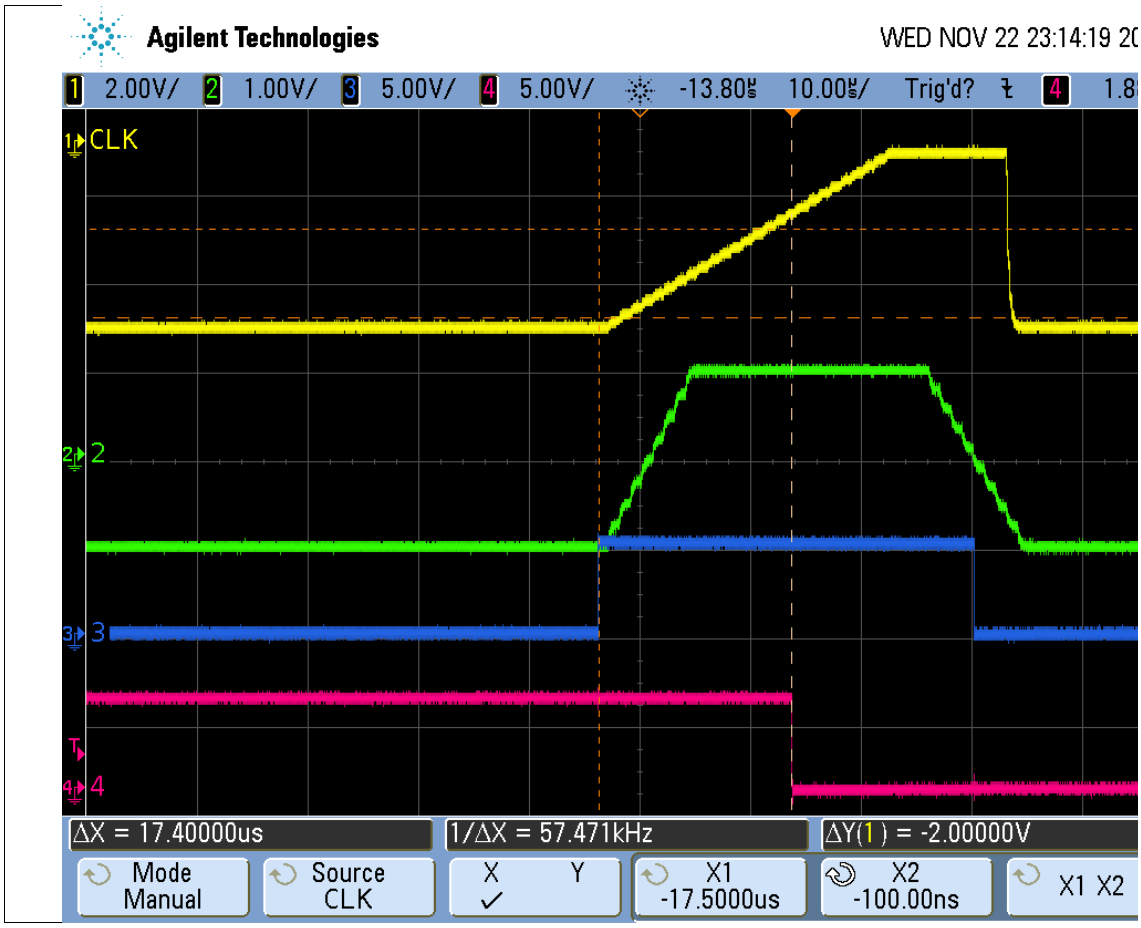
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[CIM-FUNC-0003] CIM GATE Rising state check

CIM-FUNC-0003	CIM GATE Rising state check
	PASSED

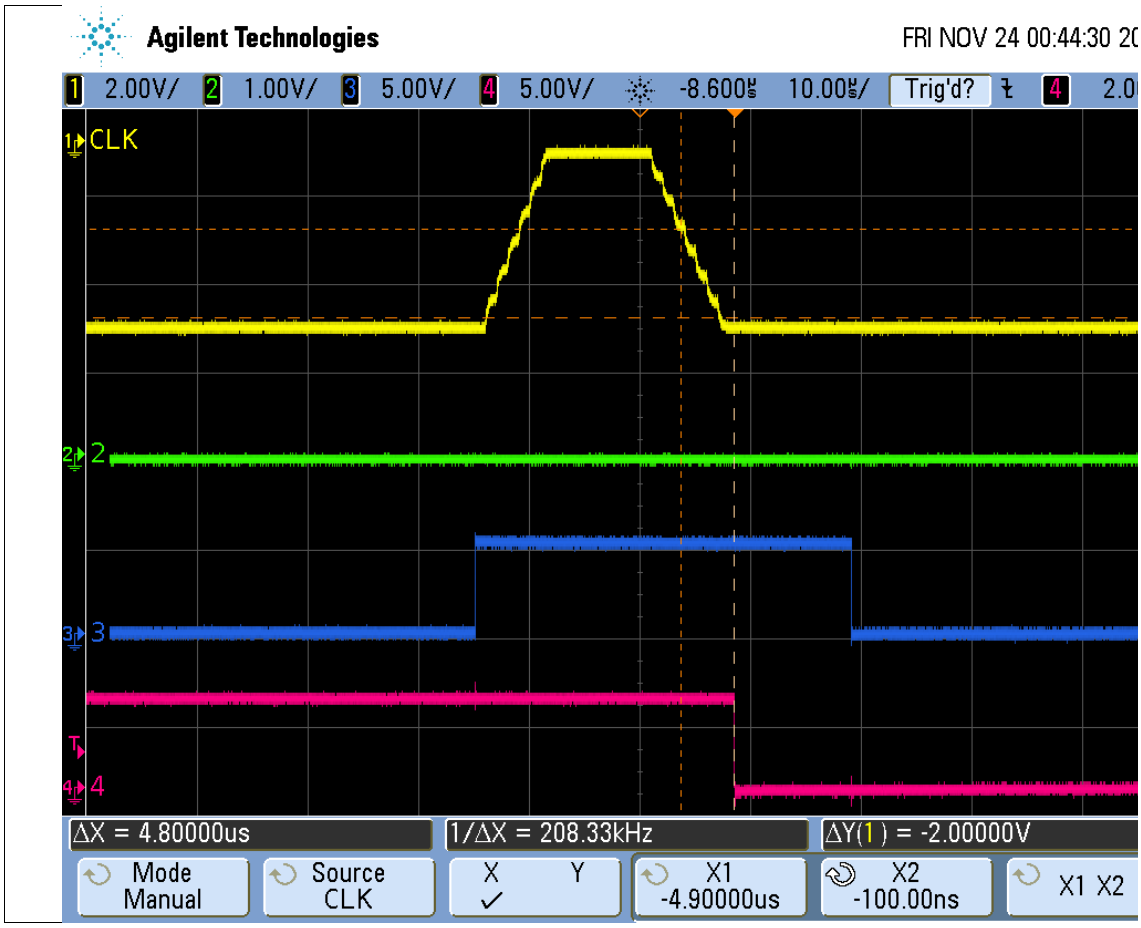
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[CIM-FUNC-0004] CIM GATE ON state check

CIM-FUNC-0004	CIM GATE ON state check
	PASSED

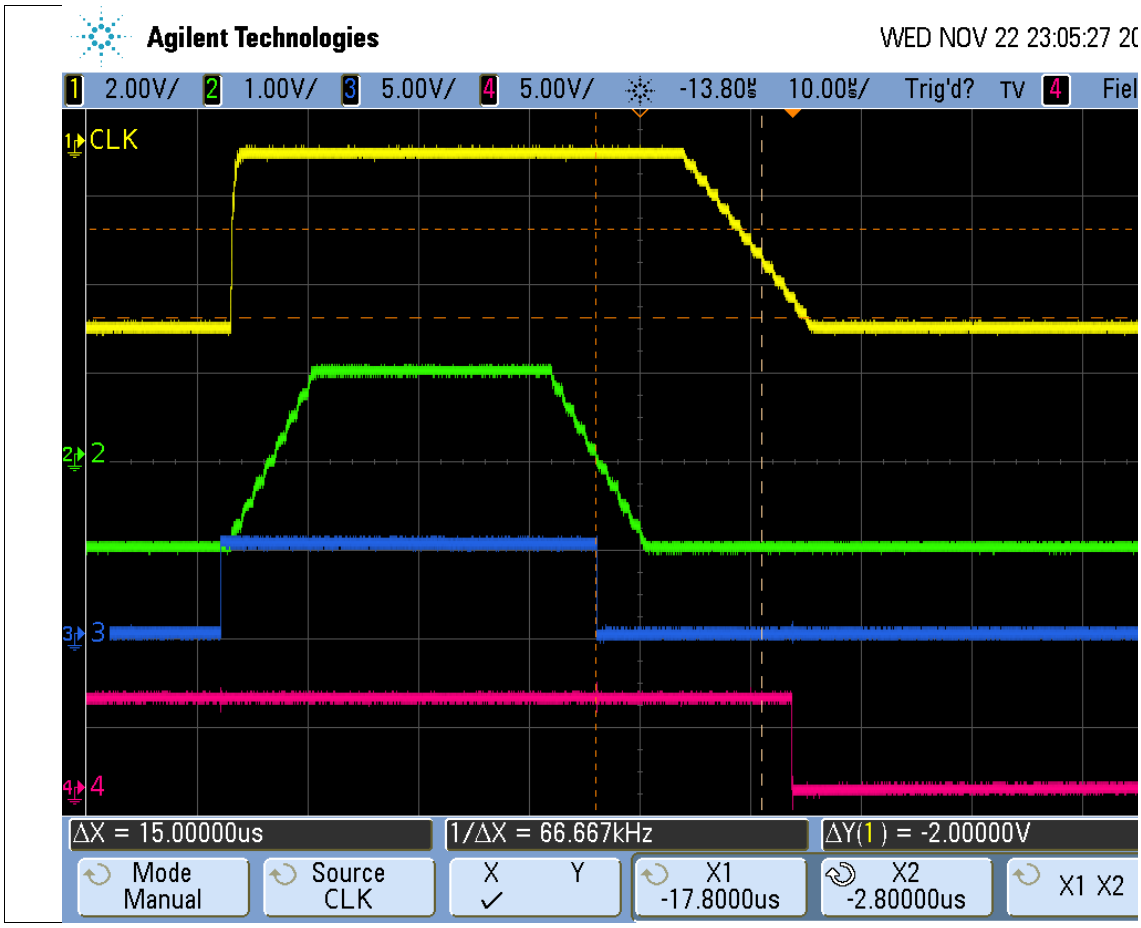
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[CIM-FUNC-0005] CIM GATE Falling state check

CIM-FUNC-0005	CIM GATE Falling state check
	PASSED

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[CIM-FUNC-0006] CIM INHIBITION

CIM-FUNC-0006	CIM INHIBITION
	PASSED

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